EXPERIMENT 1: FILE SIGNATURE ANALYSIS USING HEX WORKSHOP

AIM:

To perform file signature analysis and note the hex values of popular file types using Hex Workshop.

THEORY:

A hex editor is a tool used to view and edit binary data in hexadecimal format. It displays file contents in three sections: address (offset), hexadecimal values, and ASCII representation. Each file type has a unique file signature (magic number) at the beginning, which helps in identifying the file type regardless of its extension. Hex Workshop is widely used in digital forensics to analyze raw binary data and identify file formats.

PROCEDURE:

1. Launch Hex Workshop application.
2. Click on File → Open or use the open icon from the toolbar.
3. Browse and select a file (e.g., .docx, .jpg, .png, .pdf, .zip) and open it.
4. The file will be displayed in the editor window with three sections:
    - Offset (address) column
    - Hexadecimal values column
    - ASCII/text column
5. Observe the first few bytes in the hexadecimal column (top-left area).
6. Identify the file signature (magic number) from these initial hex values.
7. Repeat the process for different file types.
8. Record the file signatures such as:
    - DOCX/ZIP → 50 4B 03 04
    - JPEG → FF D8 FF
    - PNG → 89 50 4E 47 0D 0A 1A 0A
    - PDF → 25 50 44 46
9. Compare the signatures to understand how different file types are identified.

RESULT:

File signature analysis was successfully performed and hex values of various file types were identified using Hex Workshop.

EXPERIMENT 2: WEBSITE MIRRORING USING HTTRACK

AIM:

To use HTTrack to mirror a website by downloading its structure and content for offline browsing and analysis.

THEORY:

Website mirroring is the process of downloading all components of a website such as HTML pages, images, scripts, and other resources to create a local copy. This enables offline browsing and detailed analysis without internet access. HTTrack Website Copier is a popular open-source tool used for this purpose. It automatically crawls a website, follows links, and downloads files while maintaining the original directory structure. HTTrack supports multiple operating systems and allows users to configure options such as download depth, filters, and limits. It is widely used for website archiving, research, and forensic analysis, but must be used responsibly according to legal and ethical guidelines.

PROCEDURE:

1. Download and install HTTrack Website Copier from the official website (https://www.httrack.com).
2. Launch the HTTrack application from the start menu.
3. Click Next and create a new project.
4. Enter the Project Name, Category, and select the Base Path where the mirrored website will be stored.
5. Click Next and enter the URL of the target website to be mirrored.
6. Configure project options such as download limits, filters, or mirroring depth if required.
7. Click Next and select the action Download Website.
8. Start the mirroring process. HTTrack will crawl the website and download all related files and folders.
9. Monitor the progress of downloading through the progress window.
10. After completion, navigate to the base path and open the index.html file in a web browser to view the mirrored website offline.

RESULT:

The selected website was successfully mirrored and stored locally using HTTrack, allowing offline browsing and analysis.

EXPERIMENT 3: NETWORK PROTOCOL ANALYSIS USING WIRESHARK

AIM:

To familiarise the working of packet sniffing and analysis tool – Wireshark.

THEORY:

Wireshark is an open-source network packet analyzer used to capture and analyze network traffic in real time. It provides detailed information about packets such as source, destination, protocol, and data. It is widely used for network troubleshooting, monitoring, and security analysis. It supports multiple protocols and allows filtering, sorting, and deep inspection of packets.

PROCEDURE:

1. Open the Wireshark application.
2. Select the active network interface (e.g., Wi-Fi).
3. Click on Start Capture to begin capturing packets.
4. Observe live network traffic displayed with details like time, source, destination, protocol, and length.
5. Click on any packet to view detailed header and data information.
6. Use filters (e.g., HTTP, TCP, UDP, ICMP) to analyze specific traffic.
7. Stop the capture using the red stop button.
8. Analyze captured packets and identify different protocols.

RESULT:

Wireshark was successfully used to capture and analyze different network packets.

EXPERIMENT 4: HASH COMPARISON USING HASHCALC

AIM:

To calculate hash values using different algorithms and compare files using HashCalc.

THEORY:

HashCalc is a tool used to compute hash values (message digests) and checksums for files, text, and hex data. It supports algorithms like MD5, SHA1, SHA256, etc. A hash value is a fixed-length unique representation of data used to verify integrity. If two files produce the same hash value, they are identical; otherwise, they differ.

PROCEDURE:

1. Download and install HashCalc and open the application.
2. Select the Data Format (File, Text string, or Hex string).
3. In the Data field:
    - Browse and select a file, or
    - Enter text/hex values manually.
4. Ensure the HMAC option is unchecked for normal hash calculation.
5. Select the required algorithms (MD5, SHA1, etc.) by checking the boxes.
6. Click on Calculate or press Enter to generate hash values.
7. Repeat the same steps for another file.
8. Compare the generated hash values of both files to verify integrity.

RESULT:

Hash values were generated using different algorithms and compared successfully, verifying the integrity of the files.

EXPERIMENT 5: FAMILIARIZATION OF VARIOUS TOOLS OF THE SLEUTH KIT

AIM:

To familiarize various tools of Sleuth Kit.

THEORY:

The Sleuth Kit is a collection of command-line digital forensic tools used to analyze disk images and file systems. It allows investigators to examine data without modifying the original evidence and can recover deleted or hidden files. It supports multiple file systems such as NTFS, FAT, EXT, and others. It also includes volume analysis tools to study disk partitions. Autopsy acts as a graphical interface for Sleuth Kit, making analysis easier.

PROCEDURE:

1. Open Command Prompt.
2. Load the forensic disk image.
3. Use the following commands:
    - mmls → Display partition layout of the disk
    - fsstat → Show file system details
    - fls → List files and directories
    - ifind → Find inode number of a file
    - icat → Extract file contents using inode
4. Analyze the output of each command to understand disk structure and file details.

RESULT:

Various Sleuth Kit tools were successfully used to analyze disk image and file system data.

EXPERIMENT 7: FILE SYSTEM FORENSIC ANALYSIS – AUTOPSY

AIM:

To perform file system forensic analysis using Autopsy.

THEORY:

Autopsy is a digital forensic tool and graphical interface for The Sleuth Kit. It is used to analyze disk images and recover digital evidence without altering original data. It supports multiple file systems and provides features like file listing, keyword search, timeline analysis, metadata analysis, and recovery of deleted files. It is widely used in forensic investigations for analyzing system activity and evidence.

PROCEDURE:

1. Launch Autopsy and create a New Case.
2. Enter case name, directory, and examiner details.
3. Add a new host and proceed.
4. Select Data Source Type (Disk Image).
5. Browse and select the disk image file.
6. Set the correct time zone.
7. Configure Ingest Modules (analysis options).
8. Add the data source and click Finish.
9. Analyze files, metadata, and recovered data using available tools.

RESULT:

File system forensic analysis was successfully performed using Autopsy.

EXPERIMENT 6: FORENSIC ANALYSIS OF EVIDENTIAL IMAGE USING FTK

AIM:

To acquire a disk image and perform forensic analysis using FTK Imager.

THEORY:

FTK Imager is a digital forensic tool used to create and analyze forensic images of storage devices. It allows investigators to examine data without altering the original evidence. It supports disk imaging, file analysis, mounting images, keyword searching, and recovery of deleted data, making it useful for forensic investigations.

PROCEDURE:

1. Launch FTK Imager.
2. Click File → Create Disk Image.
3. Select the source (physical drive/removable media).
4. Choose destination to save the image.
5. Set imaging options (compression, verification).
6. Click Start to begin imaging.
7. Verify the image using hash values.
8. Click File → Add Evidence Item.
9. Load the disk image.
10. Explore files and folders.
11. View file metadata (size, date, etc.).
12. Search and extract required files.
13. Right-click the image and select Mount.
14. Assign a drive letter and mount as read-only.
15. Access the image through File Explorer.
16. Unmount the image after completion.

RESULT:

Disk image was successfully created, analyzed, and mounted using FTK Imager.

EXPERIMENT 8: FORENSIC ANALYSIS OF EVIDENTIAL IMAGE USING PRODISCOVER TOOL

AIM:

To perform forensic analysis of an image using ProDiscover and generate an investigation report.

THEORY:

ProDiscover is a digital forensic tool used to analyze disk images and recover evidence. It helps investigators examine data, detect intrusions, recover deleted files, and generate reports. It supports features like incident response, file recovery, hash analysis, and e-discovery, ensuring data integrity for legal purposes.

PROCEDURE:

1. Open ProDiscover and create a New Project.
2. Enter project name, number, and description.
3. Select Capture Image from the Action menu.
4. Choose source disk and destination path, then start imaging.
5. Load the acquired image using Open/Add Image.
6. Go to Content View to examine files.
7. Identify and recover deleted files if required.
8. Add investigator comments to selected files.
9. Copy recovered files to a folder.
10. Generate report using View → Report.

RESULT:

Forensic analysis of the image was performed and an investigation report was generated using ProDiscover.